

# EXACT 2026 - Solution Description

Team: TriAstatine · Neuro-symbolic QA · Single 8B-class LLM (Qwen3-8B + LoRA) served via vLLM (OpenAI-compatible) · Logic QA (Type 1) + Physics (Type 2).

## 1. Datasets used

- **EXACT 2026 official** - Logic-Based Educational Queries (organizer release 2026-05-15). Cleaned working set (tier-1 repair, dedup, FOL normalization, drop of logical-fallacy items): 399 records / 784 questions; fields: premises-NL, premises-FOL, question, answer, explanation. Used to train the Type 1 LoRA.
- **Physics knowledge** for Type 2: a curated formula bank covering the round-1 families; an internal physics question bank (~1.3k items) was used only to build and validate the deterministic solver. No third-party model outputs are used as labels.

## 2. Approach & method

A single FastAPI **/predict** endpoint routes by type. All LLM inference is served by a local vLLM OpenAI-compatible server; the public host also proxies **/v1/models** for model verification.

**Type 1 - Logic QA** (neuro-symbolic, 3-tier; the answer is changed only by a certified proof):

- **v40 entity-conjunctive symbolic prehandler**: a single-entity conjunctive-Horn forward chainer for the entity-grounded distribution. On multiple-choice it fires only when exactly one option is uniquely proven, returning the proof's premise indices as `premises_used`; otherwise it abstains.
- **YNU meta-unknown certificate**: fires only when the request is Yes/No/Unknown AND a premise literally states "No premise states whether/that X" AND  $\text{atom}(X)=\text{atom}(\text{question target}) \rightarrow \text{answer } \mathbf{Uncertain}$ , `premises_used` = [that premise]. Holdout-verified (779 YNU + 838 MC: 0 spurious, 0 MC-fire). Yes/No positive-proof is intentionally OFF (avoids atom-collapse false chains).
- **LoRA generation**: Qwen3-8B + LoRA (PEFT,  $r=16$ ) via vLLM `/v1/completions`; a strict parser extracts the first "Final Answer" (it does not pick stray Yes/No from the reasoning).
- **Symbolic verifier (verifier\_v35)** - rule-based, not Z3: a conservative proof checker over the natural-language premises (rules E1/PE/U1/PY). It overrides the LLM only when a rule yields a certain answer and returns the supporting premise indices; otherwise it abstains and the LLM answer is kept.
- **Answer-aware premises\_used certificate**: for a LoRA-answered multiple-choice item, `premises_used` is replaced by the proof-trace of the **returned** option only (never another option), guarded by a proof-trace source check and a length check. This improves the F1-scored `premises_used` (P2) with zero regression on the round-1 set and never changes the answer.

`premises_used` (0-based) come from the firing proof; otherwise from explicit "Premise N" citations, with a keyword-overlap fallback.

**Type 2 - Physics** (fully deterministic formula-bank solver):

A self-contained family classifier  $\rightarrow$  formula bank covering **electricity** (Ohm/power, series and parallel resistance/current, capacitor energy and series charge, Coulomb force, electric-field superposition, electric potential, resistivity  $R=\rho \cdot l/S$ ,  $R=P/I^2$ , LC resonance, inductive reactance), **mechanics** (kinematics, work-energy final speed, apparent weight in an accelerating elevator  $N=m(g \pm a)$ , projectile height), **thermodynamics** (calorimetry  $m \cdot c \cdot dT$ , latent heat  $m \cdot L$ , ideal gas  $nRT/V$ ), **transformer** turns ratio, and **thin-lens** optics. A strict SI unit parser (prefixes p/n/u/m/c/k/M/G, cm, mm, min- $\rightarrow$ s) feeds the formulas.

**Output discipline (key)**: the grader compares the value with a numeric tolerance but matches the **unit string** strictly, so the solver emits the grader-canonical unit (e.g. N/C not V/m, Pa not MPa, J not kJ; uF / uC / uJ where expected) and the value as a **plain ASCII decimal** (no scientific 'x 10^' notation). Families outside the bank fall back to the same Qwen3-8B (no second model). On the round-1 evaluation set all 25 Type-2 items are solved with canonical units, and the deterministic route does not fire on out-of-bank physics. `premises_used` = [].

## 3. Reliability

A strict full-endpoint gate (local and public: `/health`, `/v1/models`, `/v1/completions`, `/predict` for each Type-1 tier and for Type-2, plus the error-body route) must pass before the server is used. Type-1 predictions are additionally validated by a reload-and-recompute safety step and a conservative selector, so symbolic corrections are applied only when proof-certified.

## 4. Model-size calculation (within 8B-class)

- Exactly one LLM is loaded and running: Qwen3-8B (~8.2B total parameters); Type 1 adds a small LoRA adapter ( $r=16$ ), and the Type-2 fallback uses the same base. No second LLM is co-resident, so total active parameters stay within the 8B-class limit.
- Non-LLM components - the rule-based symbolic verifier and the deterministic physics solver - do not count toward the limit.

## 5. Serving & verification (vLLM)

- vLLM OpenAI server (tensor-parallel across 2xT4, TRITON attention backend for Turing) exposes `/v1/models` reporting the declared ids `qwen3-8b-base` and the LoRA `qwen3-8b-exact-type1`.
- The public Cloudflare URL serves `/predict` and proxies `/v1/models` and `/v1/completions` (all listed in `urls.txt`). No third-party inference APIs; all served weights are open-source.